

Towards a Virtual Reality Math Game for Learning In Schools - A User Study

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ABSTRACT

In recent years, immersive Virtual Reality (VR) has gained popularity among young users as a new technology for entertainment gaming. While VR remains majorly used for entertainment purposes, 3D desktop games are already used in schools. This study takes a closer look at the suitability for VR games to be used in a formal educational environment, and its potential to enrich existing game based learning approaches. Based on learning needs of in particular easily distracted and inattentive children, an immersive VR math game was created and tested on 15 children aged 11-12.

Keywords: Immersive virtual reality, game based learning, applied immersive gaming, serious games

1 INTRODUCTION

In the past two decades, games have been increasingly explored as new tool in education. Educational games fall under the umbrella of serious games, games designed for a purpose other than pure entertainment [1]. Researchers such as Young et al. [2] found that games are an effective tool for knowledge acquisition and foster a positive student attitude [3]. Kirriemuir and McFarlane [4] indicate that video games may improve communication skills, as well as strategic thinking abilities and planning skills. Despite these promising findings, several studies suggest no significant positive effect of desktop video games for learning [5], as well as other methods of instruction in formal education being as effective as video games [6]. This poses the question how video games may be best implemented in education for maximum learning effectiveness. This can be looked at from different application fields such as pedagogy, educational science, or technology.

Furthermore, the question in research remains to what extent games may be adjustable to target specific learning needs, increasing learning effectiveness. Lastly, little research exists on exploring other new technologies for serious gaming, such as immersive VR, and their potential to improve and enrich learning effectiveness of gaming in a formal education context.

Immersive VR technology has become increasingly accessible and usable. VR has raised interest in serious game research due to its promising effects on learning. For example, VR seems to motivate users, supports information-retention through immersive visualization [7, 8].

This study explores the implementability of VR games for children in formal education, whereby the math game designed in this research factors in design requirements for children challenged by attention deficits and hyperactivity, making it an inclusive approach. The math game was tested on 15 children aged 11-12 in a school environment, resulting in valuable feedback on its acceptance in an educational setting, and game design elements for immersive VR learning games designed inclusively.

2 THEORY

As an extensive literature review could not reveal guidelines for serious educational VR game design for children, game elements for the creation of the math game in this study were constructed in an interdisciplinary manner. Literature on existing VR games and their respective elements, mitigation strategies for unique learning needs in schools, and elements from educational science were taken into account. This was accompanied by a situation analysis of formal educational school contexts. In the following, an overview of the defined VR game design requirements is portrayed.

2.1 Design Requirements

To help children cope with a demanding school environment, the look and feel of the VR environment should be welcoming, safe and aesthetic, accommodating for all learners [9]. Since VR is a rather new and often unfamiliar technology [7], a training scenario shall precede the game play, helping users to acquaint themselves with the technology and game elements, supporting user acceptance. As VR allows for active movement, which in turn has been linked to better learning outcomes especially in learners with hyperactivity [10], one design requirement for the game is actively incorporating body movement. To support guiding behavior and actions, clear verbal cues in task-switching situations are required [11]. Moreover, to stimulate working memory and attention of learner, only few information at a time shall be displayed, comparing distraction and information overload [12,13]. To further assist children that display self-regulation difficulties, no imposition of time limits should be used, reducing the risk of negative arousal [12]. Lastly, a game design requirement for this study is to create an environment that allows enough time to observe the task as well as visual cues, with no time impositions, and no not needed visual cues that could mislead or confuse the user [14].

3 USER STUDY

For the user study, an immersive VR math game was created, aiming at identifying the applicability of immersive VR in a school context, as well as defining game requirements for such a game. The game was created based on defined requirements and comprises a training room scenario and a gameplay scenario. In the training room, users acquire navigational and technical skills needed for when they proceed into the game room. The game itself comprises of a clean room with the look and feel of a room in space, where warm colors contribute to a calm atmosphere. The user can find 4 numbered cubes and 4 balls each displaying operators on the ground. The task is to pick up the items and throw them into dedicated snap zones, in the right order to meet a

predefined target number. This is essentially the practice of basic equation solving.

Through existing networks, a school in New Zealand was recruited for this study. A total number of 15 children, whereby 14 were male and 1 was female, participated in the user test which was conducted during normal school hours inside the educational facility.

3.1 Method

The user test consisted of two different parts: A usability game test, and a post-test user experience evaluation. During the usability test, each child was asked to play the game for 10 minutes, whereby observations were taken and the screen was recorded. For the experience evaluation, tools were used to gather opinions in child computer interaction. Additionally, the children were each asked 6 semi-structured interview questions which were audio recorded. This mixed-method approach was chosen to receive a variety of self-reported and observed data.

3.1 Results

The average time for completing the training in the respective training room was 3:12 min, whereby 14 out of 15 participants demonstrated a fast progression through the rooms (> 5 min). All participants reported to like the training rooms, while 5 ranked them highest on the scale, as brilliant. Out of 15, 13 participants would train in the room again.

All 15 participants indicated that they liked the game, while 7 found it really good, and 5 brilliant. Moreover, all children reported to at least feel good during the training and gameplay. Further, 14 children would play the game again, and would throw the cubes and balls again. Here, 5 participants reported this to be their favorite part. All 15 participants would use VR glasses again. Other parts enjoyed were learning and figuring out mechanics and controls, the freedom to explore, figuring out the math, finishing a question, the game's simplicity, the environment, as well as receiving guidance from the facilitator. Parts not enjoyed were reported by 3 participants and include navigating the space, doing math, and throwing. Based on self-report, 12 children were familiar with VR. It was observed that 13 children moved around rather less while being in VR, while also 13 children were confident with the technology. In total, 3 children had questions about technology, and 14 participants handled the technology intuitively. Out of 15 children, 3 purposefully moved outside the defined play area/stationary boundary to explore outside of the defined VR space. In all 3 cases this led to distractions for the user. Further, 3 participants completed the first equation in the game, while all 3 individuals appeared engaged and motivated by their success. It was observed that 4 children talked out loud while thinking how to solve the math equation. On the other hand, 3 out of 15 participants did not talk at all, even when asked a question. This however did not compromise intense focus and concentration, all 15 participants stayed on task. Lastly, 6 children self-reported that they are having fun and/or enjoy VR. No child reported symptoms of VR motion sickness.

3.2 Discussion and Conclusion

It can be concluded that most children in this study enjoyed the usage of VR glasses, as well as training VR controls and playing the game, indicating that VR may be able to become an accepted learning medium for children. Moreover, the VR setup and gameplay was well implementable into the daily school routine and did not require special space considerations as most participants did not move much. This also underlines its potential applicability as learning tool in educational environments. In

terms of game design, a simple and clean scene with no time impositions seems to be effective in keeping participants on task. VR seemed to motivate children to complete the task and lead overall to enjoyment and the feeling of having fun. This is congruent with former research. Most children had prior experience with VR, and 14 did well in the training rooms preparing them for the mechanics of the game play. This may indicate that clear verbal cues may assist control and mechanics learning. No child had major issues with the technology and mostly handled controls intuitively. Hence, barriers in terms of technical knowledge and handling were not identified. In this study, it appears that the throwing of objects (cubes and balls) was enjoyed by the majority of children. This is underlined by the fact that throwing cubes and balls was mentioned most frequently as most fun. Hence, active movement appears to be enjoyed and marks a unique game element of VR. For future studies, customization options for verbal information retention as well as physical movement could be added. One participant for example would have favored orally communicated instructions. Lastly, the gender balance in this study must be seen as a limitation. While 14 out of 15 children were of the male gender, only 1 participant can be identified as female. This implies the need for further testing this technology on both genders equally.

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